IS1212 : Probability and Statistics

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Topics to be covered

- 1: Introduction to Statistics and basic concepts
- 2 : Measures of location and dispersion
- 3 : Introduction to Probability and Probability theories
- 4 : Discrete probability distributions
- 5 : Continuous probability distributions

Introduction to Statistics

Chapter 01

What is Statistics?

The science of

collecting

organizing

analyzing

presenting

data,

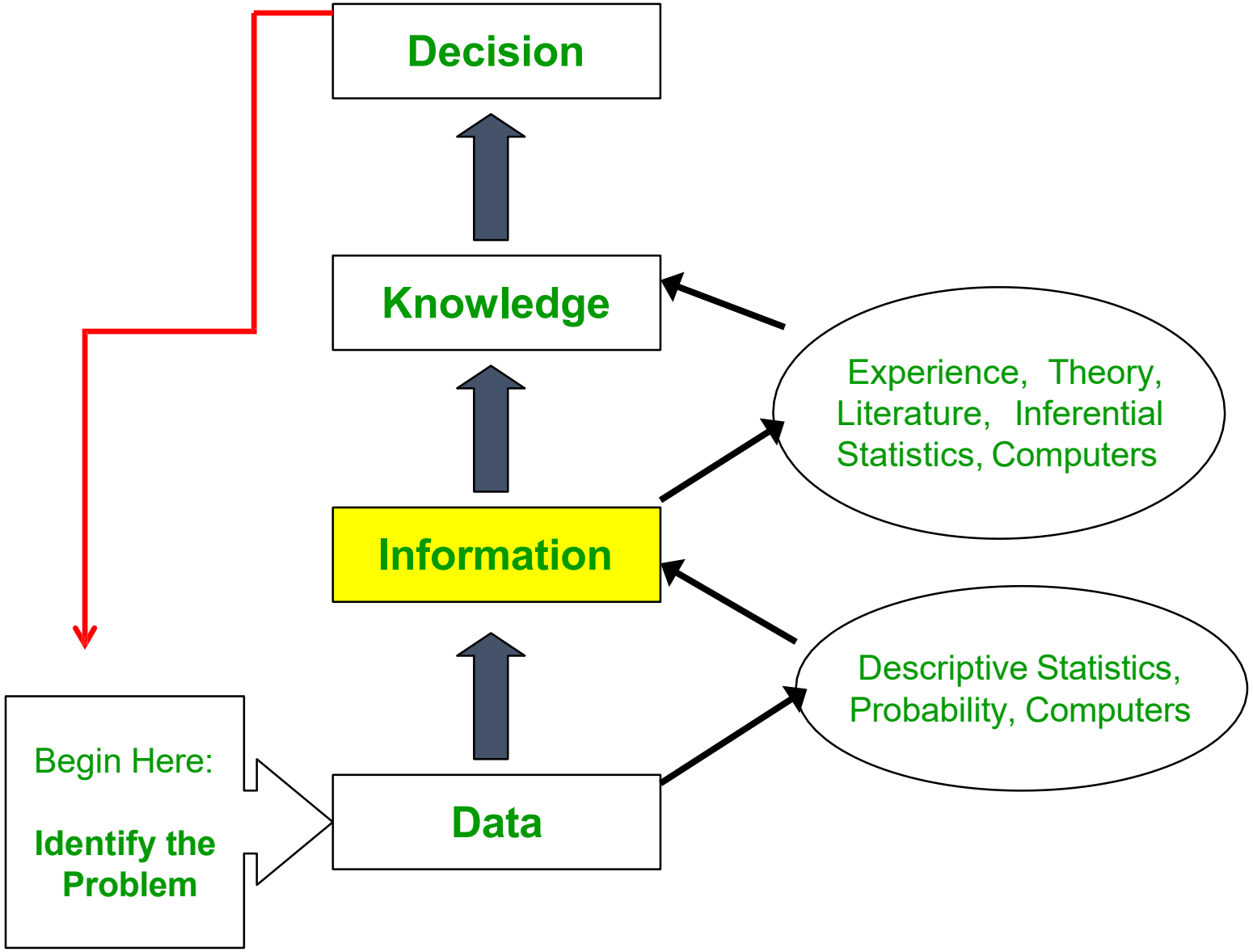
and

finally drawing conclusions

in the

best possible way

Use of Statistics in Decision Making



Basic Terms

- **Population**
  - the complete collection of individuals or objects that are of interest to study.
- **Sample**
  - a subset of the population.
- **Variable**
  - observable, measurable or recordable characteristic.
  - basic unit of analysis.
- **Observation**
  - a value that a variable assumes for a single element

Basic Terms…

- **Data**
  - The set of observations collected for the variable
- **Parameter**
  - a numerical summary measure used to describe a characteristic of a population.
- **Statistic**
  - a numerical summary measure used to describe a characteristic of a sample.

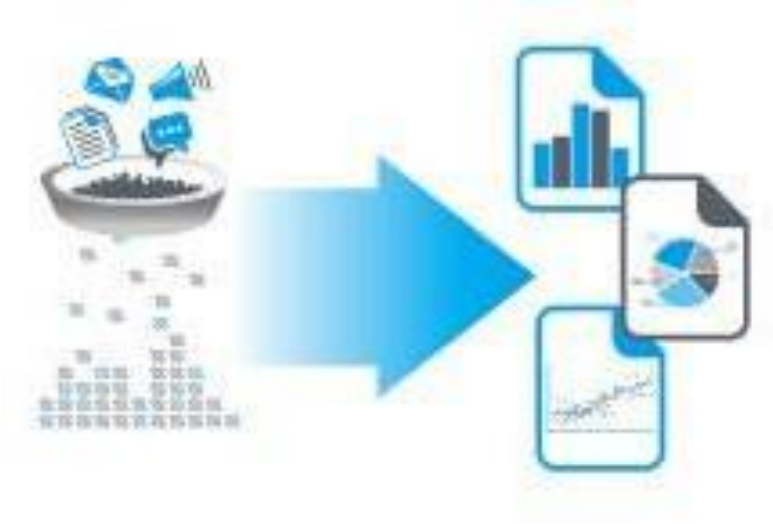
Statistical Software

EXCEL, SPSS, MINITAB, SAS, S Plus, Matlab, R, Python, and more…

What is Data Analysis?

- **Data**?
  - Raw facts
  - Observations, Measurements, Answers

- **Analysis**?
  - Extracting information



- **Data Analysis**?
  - The process of extracting useful information from the data

Quantitative vs Qualitative

Qualitative Data Analysis

- Identifying themes
- Identifying patterns
- Describing situations

Quantitative Data Analysis

- Use numerical summaries
- Creating Codes to define categories
- Counting instances to see frequency
- Coding and Recoding to see ranges
- Making comparisons between groups
- Presenting findings in tables and graphs
- Deepening analysis with **statistics**

Why We Study Statistics?

- To evaluate published data
- To collect/summarize data meaningfully
- To present the findings (reports or verbally)
- To interpret the statistical results
- To employ statistical methods to make decisions
- To develop new techniques
- To generalize the decisions

The Main Functions in Statistics

- Formulate a real problem in statistical terms.
- Collect data efficiently
- Analyze data efficiently and extract the maximum amount of information.
- Interpret and report the results.

- Need to have sufficient background knowledge and a range of statistical techniques.
- The key features of statistics is to making decisions about a population in the presence of variability between the individuals.

Examples

1. Agriculture

   To find out the main factors associated with the yield.

2. Medicine

   To compare a new drug with a standard drug

3. Marketing

   To find out approximate demand for a product

4. Computer Science

   To measure the efficiency of computer systems

Summarizing Data

Summarizing Data

- In practice we have huge amount of data.
- It is difficult to look at each observation and draw conclusions.
- Need ways to summarize the data to bring out their main features.

Descriptive Statistics

- Deals with methods of describing some features of data, without making any conclusions.

- Descriptive statistics, is an important area of statistical analysis.
    - To show the main features of the data.
    - To make tentative decisions from the data.

Inferential Statistics

- Inferential statistics would concern generalizing from a sample to the population from which it is selected.

- We make some decisions (inferences) using procedures, where certain assumptions are made about the data.

- It deals with methods that enable a decisions or inference to be drawn from analysis of data.

Types of Variables
and
Levels of Measurements

Variables

- Characteristics of interest, to study.
- Different statistical methods are appropriate for the different type of variables.
- There are two kinds of variables.
  - Numerical (Quantitative) Variables
  - Categorical (Qualitative) Variable

Categorical (Qualitative) Variable

- Variable having categories or classifications that are not numerical in nature.
  - Examples
    - Gender **(Male, Female)**
    - Social class of a person **(High, Med, Low)**
    - Quality of computers **(Good, Bad)**
    - Your hair color
    - Your ethnicity
    - Did you pay income tax last tax year?

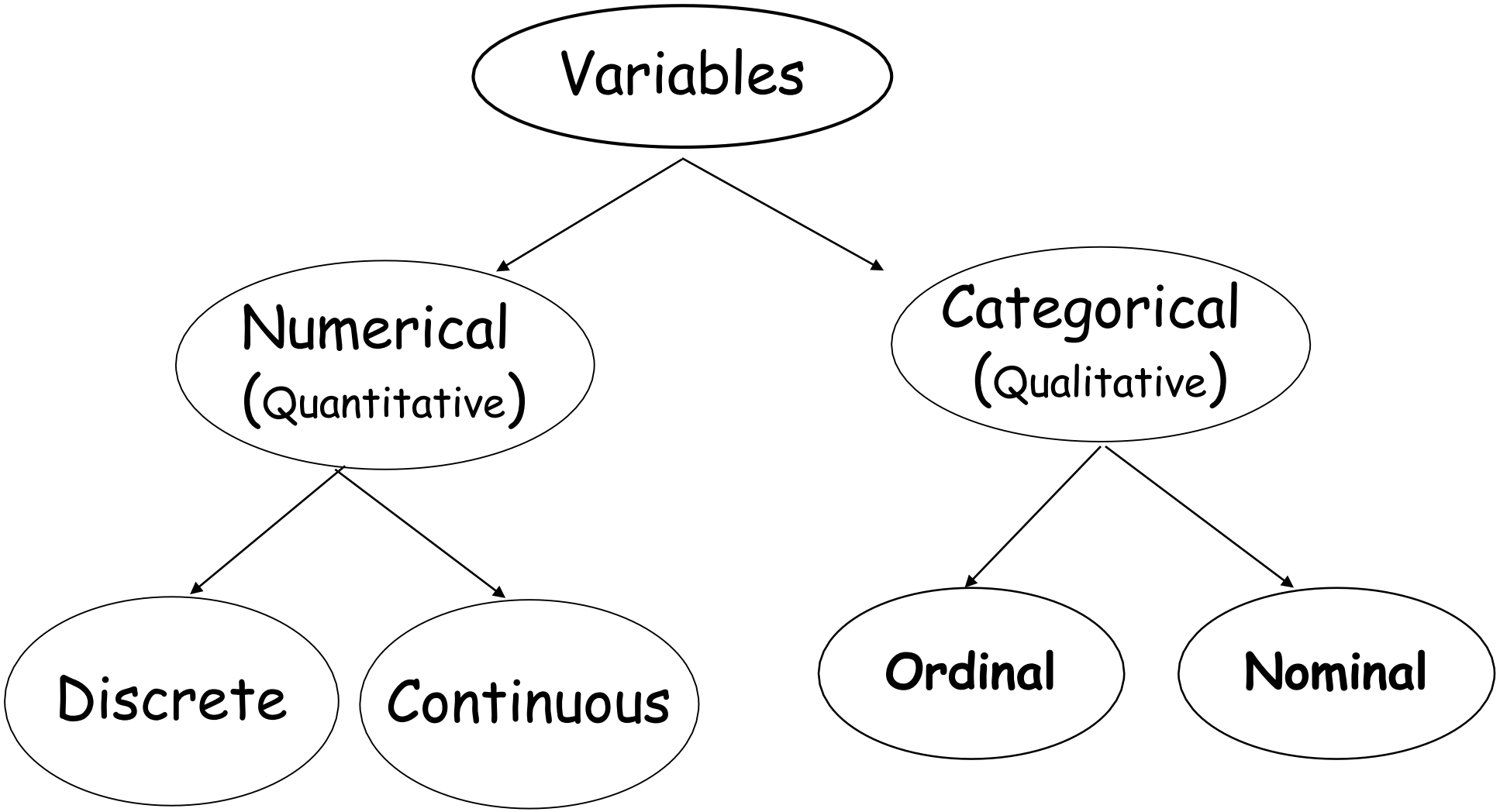
Numerical (Quantitative) Variable

- Variable whose values are numerical in nature.
    - Weight, Height, Volume,…
    - Exam marks
    - Number of defective parts in a computer
    - Age of a person
    - Blood cholesterol level a person
- Quantitative variable can be further sub divided as, **Discrete** Variable and **Continuous** Variable

Discrete and Continuous Variables…

- **Discrete variable**
  - **There is a gap between two possible values**
  - take only countable or finite values.
    - Number of customers arriving at a supermarket.
    - Number of defective parts in a computer.
- **Continuous variable**
  - ***Theoretically*, no gap between two possible values**
  - take uncountable number of values or any real values.
    - Amount of rainfall.
    - Time taken to complete a computer job.

Types of Variables



Primary and Secondary Data

- **Primary Data**
  - Data collected for a specific purpose.
  - Data collected at first hand.
  - It involves measuring, taking readings, sketching, counting, or interviewing (using questionnaires).

- **Secondary Data**
  - Data which were collected for another purpose.
  - Data collected from external sources
    - TV, radio, internet, magazines, newspapers, research articles, stories told by people you know

Statistical Studies

and

Sampling

Statistical Studies

Basically, there are two types of studies.

- **Observational Study**

  In observational study the relevant data are collected without controlling other factors.

  eg: In order to study the factors that may affect on employees' productivity, a random sample of employees is selected and recorded their education qualification and experience.

- **Experimental Study**

  In Experimental study the relevant data are collected by controlling some other factors. That is, the study is under the control of the experimenter

  Eg: A certain chemical experiment can be conducted by controlling pressure and temperature.

Types of experimental studies

- **Comparative Experiment**

An experiment conducted to compare two or more populations is called Comparative experiment.

Eg: To compare effect of two drugs produced for the treatment of headaches.

Compare the sales of a food item depending on the type of package design.

Note: As a general rule, when comparing treatments, it is wise to include a control group, if we don't know at least one treatment is significantly different from the control (no treatment). Then the comparisons among treatments have more practical use.

- **Absolute Experiment**

An experiment conducted to determine the value of some characteristic of the population.

Eg: If data are collected to find the average age which you can harvest a certain crop under controlled environment.

Sampling

- The process of taking a sub-set from a population
- For a good sampling need to:
  - Specify the objectives of the study clearly
  - Identify the variable or variables of interest
  - Select the appropriate sampling technique
  - Collect correct data

- Representative sample:

  Exhibits characteristics similar to the population

Probabilistic Sampling

- **Simple Random Sampling**
  - The probability of being selected is "known and equal" for all members of the population
- **Systematic Sampling**
  - Select every kth element from a randomly selected item
- **Stratified Sampling**
  - The population is separated into homogeneous groups (strata) and a random sample is taken from each strata. The samples are then combined together.
- **Cluster sampling**
  - The population is divided into groups (clusters) and a random sample of clusters is taken.

Non-probabilistic Sampling

- **Convenience Sampling Method**
  - Samples drawn at the convenience of the interviewer. People tend to make the selection at familiar locations and to choose respondents who are like themselves.
- **Purposive Sampling**
  - Purposely select a sample to confirm predefined "idea"
- **Judgment Sampling Method**
  - Samples that require a judgment or an "educated guess" on the part of the interviewer as to who should represent the population.
- **Snowball Sampling Method**
  - Identify the 1st person. Then find the 2nd item through the information received from 1st person. 3rd person is find through the information received from 2nd person and continue until get a satisfactory sample.
- **Quota Sampling Methods**
  - Samples that set a specific number of certain types of individuals.

Any questions?

Answers?

1. Why Computer Science needs Statistics?
2. What is the difference between parameter and statistic?
3. Indicate the type of these variables: Religion, Name, GPA, Grade, Temperature, Number of Students in a Classroom
4. Ashan collected data from university records for his study. What type of data source Ashan has accessed?
5. "**ABC school wants to know the average height of its students. There are 800 students in total. The school measures the height of 50 randomly chosen students.**" Identify the population and sample for this scenario.
6. Provide two examples for both probabilistic and non-probabilistic sampling approaches.